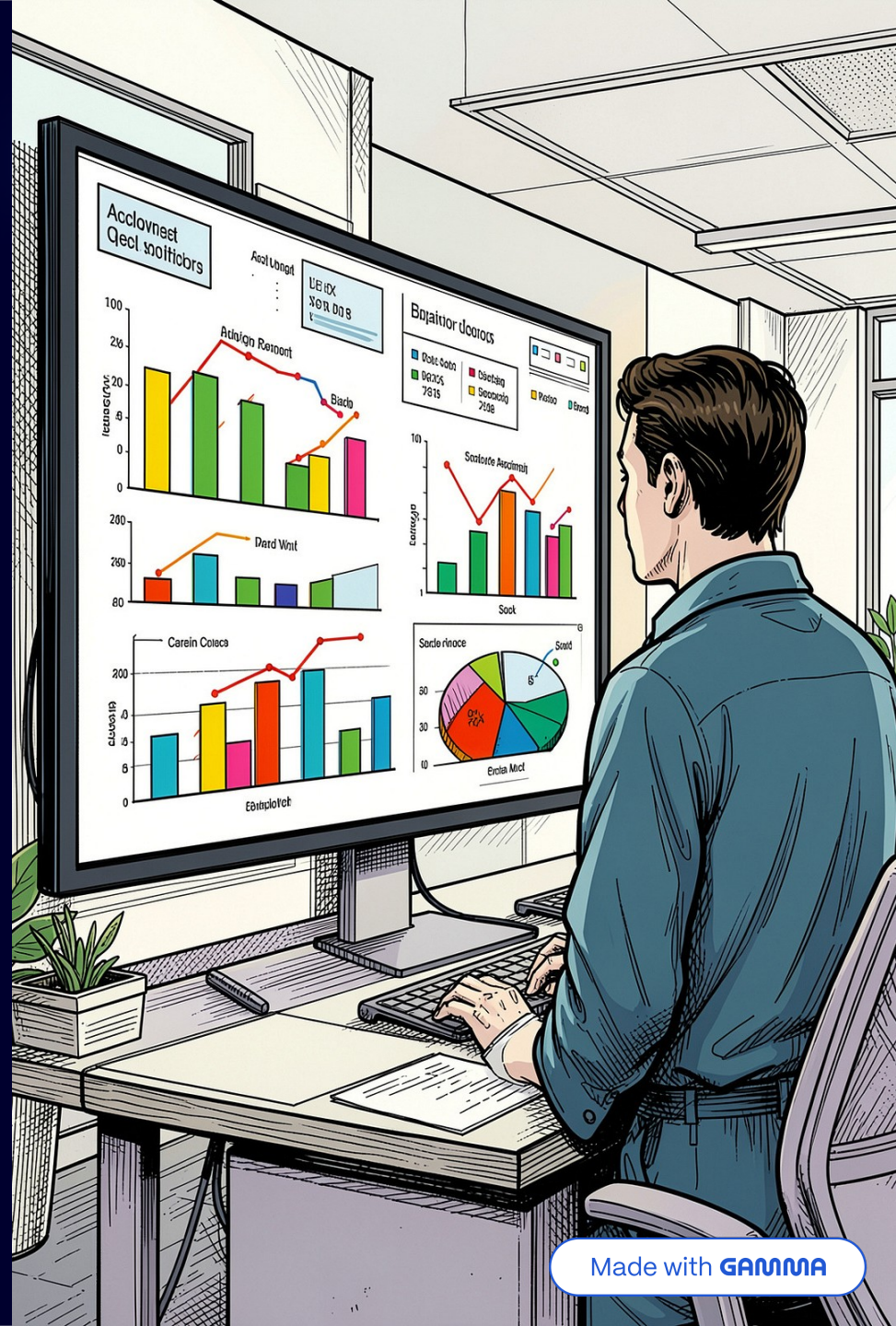
Data Visualization with Plotly



What is Data Visualization?

Data Visualization is the process of representing data visually using charts and graphs.

Instead of looking at a large table of numbers, we use visualization to make patterns, relationships, comparisons, trends, and distributions easier to see and understand.



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Why is Data Visualization Important?

Why don't we just look at the data in the table?

A table is useful when we need to find a specific value, but it is not always easy to understand the overall picture.

Visualization helps us quickly identify:

- Patterns
- Trends
- Relationships
- Differences
- Distributions
- Outliers

For example, if we have hundreds of exam scores, looking at the table can tell us the individual scores, but a chart can quickly show us whether most students have low, medium, or high scores.

The goal of visualization is not just to make data look better. It is to make data easier to understand.

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Types of Data Analysis

Univariate Analysis

Uni = One

Analysis of one variable.

We focus on understanding the distribution or characteristics of a single variable.

Examples:

- exam_scores
- age
- monthly_salary

Example: exam_scores

We may ask: How are the exam scores distributed?

Bivariate Analysis

Bi = Two

Analysis of two variables.

We usually look for a relationship, comparison, or difference between them.

Examples:

- study_hours & exam_scores
- department & salary

Example: study_hours & exam_scores

We may ask: Is there a relationship between study hours and exam scores?

Multivariate Analysis

Multi = More Than Two

Analysis of three or more variables.

We analyze multiple variables together to understand more complex relationships or comparisons.

Examples:

- study_hours & exam_scores & grade
- department & gender & salary

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Important Charts

The main charts we will use are:



Histogram



Bar Chart



Pie Chart



Box Plot



Line Chart



Scatter Plot

The important question is: **When should I use each chart?**

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Histogram

Used for:

Numerical Data → Distribution → Univariate

A Histogram shows how numerical values are distributed across different ranges.

Key idea: Histogram = Numerical Data + Distribution

Example: exam_scores

Question: How are exam scores distributed?

Another example: monthly_salary

Question: What is the distribution of monthly salaries?

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Box Plot

Used for:

Numerical Data → Distribution + Five-Number Summary → Univariate

A Box Plot gives us a compact summary of the distribution.

It represents the:

- Minimum
- Q1
- Median
- Q3
- Maximum

It can also help identify outliers.

Key idea: Box Plot = Distribution + Five-Number Summary + Outliers

Example: monthly_salary

Question: What is the salary distribution and are there any outliers?

Another example: exam_scores

Question: What is the distribution of exam scores?

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Bar Chart

Bar Charts are mainly used with categorical data.

Univariate

Categorical Data

Example: department

Question: How many employees are in each department?

Bivariate

Categorical Data + Numerical Data

Example: department & average_salary

Question: What is the average salary for each department?

Multivariate

Categorical Data + Categorical Data + Numerical Data

Example: department & gender & average_salary

We can use barmode to compare categories. For example: department & gender & average_salary with barmode="group" or barmode="stack"

Key idea: Bar Chart = Comparing Categories

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Pie Chart

Univariate

Categorical Data

Used to show the proportion of categories.

Example: department

Question: What percentage of employees belong to each department?

Bivariate

Categorical Data + Numerical Data

Example: department & employee_count

Question: What percentage of employees belong to each department?

Pie Charts work best when there are only a few categories, usually around 3–4 categories.

Key idea: Pie Chart = Proportions of Categories

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Line Chart

Used for:

Time Series Data

Usually:

- Time → X-axis
- Numerical Value → Y-axis

Line Charts are mainly used to show trends over time.

Key idea: Line Chart = Change / Trend Over Time

Example: date & sales

Question: How did sales change over time?

Another example: month & temperature

Question: How did temperature change throughout the year?

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Scatter Plot

Used to understand the relationship between two numerical variables.

Bivariate

Numerical Data + Numerical Data

Example: study_hours & exam_scores

Question: Is there a relationship between study hours and exam scores?

Multivariate

Numerical Data + Numerical Data + Categorical Data

Example: study_hours & exam_scores & grade

We can add another variable using properties such as:

color, size, symbol

Key idea: Scatter Plot = Relationship Between Numerical Variables.

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Choosing the Chart

Think about the question first.

Distribution of numerical data

→ Histogram

Distribution + Five-Number Summary

→ Box Plot

Compare categories

→ Bar Chart

Show proportions of a few categories

→ Pie Chart

Show trends over time

→ Line Chart

Show relationship between numerical variables

→ Scatter Plot

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Plotly

Plotly is a Python library used to create interactive visualizations.

Basic structure:

```
import plotly.express as px
```

Then we create the chart using functions such as:



Histogram with Plotly

```
px.histogram() px.bar() px.pie() px.box() px.line() px.scatter()
```

```
fig = px.histogram(
    df, x="exam_score", title="Exam Score Distribution"
)

fig.show()
```

Important Parameters

x

The numerical column we want to visualize.

title

The title of the chart.

nbins

Controls the number of bins.

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Bar Chart with Plotly

```
px.histogram(df, x="exam_score", nbins=10)
```

```
fig = px.bar(
    df, x="department", y="salary", title="Average Salary by Department"
)

fig.show()
```

Important Parameters

x
The categorical variable.

y
The numerical value.

text_auto
Displays values directly on the bars.

`text_auto=True`

color
Adds color based on a variable.

barmode
Controls how multiple categories are displayed.

`barmode="group"` or `barmode="stack"`

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Pie Chart with Plotly

```
fig = px.pie(
    df, names="department", values="employee_count", title="Employees by Department"
)

fig.show()
```

Important Parameters

names

The categorical variable.

values

The numerical value or count.

color

Controls the colors of categories.

color_discrete_sequence

Controls the color sequence for categorical data.

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Box Plot with Plotly

```
fig = px.box(
    df, y="salary", title="Salary Distribution"
)

fig.show()
```

For comparing distributions between categories:

```
fig = px.box(
    df, x="department", y="salary", title="Salary Distribution by Department"
)

fig.show()
```

Important Parameters

x

Categorical variable when comparing groups.

y

Numerical variable.

color

Adds another categorical dimension.

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Line Chart with Plotly

```
fig = px.line(
    df, x="date", y="sales", title="Sales Over Time"
)

fig.show()
```

Important Parameters

x

Usually the time variable.

y

The numerical value.

color

Creates separate lines for different categories.

Example:

```
px.line(
 df, x="date", y="sales", color="product"
)
```

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Scatter Plot with Plotly

```
fig = px.scatter(
    df, x="study_hours", y="exam_scores", title="Study Hours vs Exam Scores"
)

fig.show()
```

Important Parameters

x
First numerical variable.

y
Second numerical variable.

color
Represents a categorical or numerical variable using color.

size
Represents another numerical variable using marker size.

symbol
Represents a categorical variable using different marker shapes.

Example:

```
fig = px.scatter(
 df, x="study_hours", y="exam_scores",
 color="grade", size="attendance", symbol="gender"
)
```

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Common Plotly Parameters

These parameters can be used across different charts.

title

Adds a title.

`title="Sales Analysis"`

labels

Changes axis and legend labels.

```
labels={
    "exam_score": "Exam Score",
    "study_hours": "Study Hours"
}
```

color

Adds another dimension using color.

`color="grade"`

template

Changes the overall visual theme.

`template="plotly_dark"`

text_auto

Displays values directly on the chart.

`text_auto=True`

markers

Used with charts such as Line Charts to display individual points.

`markers=True`

color_discrete_sequence

Controls colors for categorical variables.

`color_discrete_sequence=px.colors.qualitative.Set2`

color_continuous_scale

Controls the color scale for numerical variables.

`color_continuous_scale="Viridis"`

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Final Mental Model

Before creating any chart, ask yourself three questions:

1. **What type of data do I have?**

   Numerical or Categorical?

2. **How many variables am I analyzing?**

   Univariate, Bivariate, or Multivariate?

3. **What do I want to understand?**

   Distribution, Comparison, Relationship, Proportion, or Trend?

Then choose the chart.

Data → Question → Analysis Type → Chart → Insight

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